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Research Paper

A composite laser ablation diagnosis method based on multiple spectroscopic and imaging analyses

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ABSTRACT

This study proposes a novel diagnostic approach for target ablation to comprehensively elucidate the physical mechanisms of laser ablation in aluminium alloys and stainless steel, precisely measure sample temperatures, and predict the ablation state. The method utilizes a spatially weighted emissivity model in conjunction with multispectral thermometry techniques to analyze spatial variations in temperature and emissivity distributions, facilitating the evaluation of target ablation status. Through a series of experiments, temperature data obtained using an enhanced weighted radiative spectral inversion technique were compared with temperatures recorded by thermal imaging cameras, confirming the effectiveness and accuracy of the weighted radiative spectral inversion method in multispectral thermometry. Additionally, a detailed examination of laser ablation in aluminium alloys and stainless steel was conducted to elucidate the underlying damage mechanisms. This refined approach establishes a solid groundwork for further investigation into the characteristics and dynamic Evolution of laser-damaged regions.

1. Introduction

Multispectral thermometry, a highly accurate non-contact temperature measurement technique [1,2], demonstrates significant application value in diverse fields such as materials science research, aerospace engineering, and military strategies [3–6]. Particularly in laser damage assessment, this technology excels in real-time monitoring of target surface temperatures, precise quantification of material emissivity [7,8], and maintaining adaptability in various environments. These capabilities significantly improve the analysis and prediction of material damage mechanisms, providing deep insights into the underlying processes of laser-induced damage. The spectral radiation measurement range is limited for target damage and its surrounding areas during laser irradiation. However, the relatively uniform temperature distribution within the damaged region allows for precise assessment of thermal conditions in these dynamically changing zones using multispectral temperature measurement techniques, making it a valuable tool for temperature assessment in laser-induced damage.

Recent research in multispectral thermometry for laser damage assessment has made significant progress, with a focus on enhancing measurement accuracy [9], improving environmental adaptability, and facilitating practical applications of the technology [10,11]. Methods for temperature inversion in laser-damaged areas using multispectral thermometry are continuously being refined and enhanced [12–14]. Yalou Qin et al. proposed an enhanced spectral thermometry method to investigate alterations in the damaged area's condition and temperature distribution following laser-induced damage. Through the application of this method for temperature inversion on the radiant spectra of the affected region, they achieved a minimum 24 % reduction in temperature measurement errors [15]. Lei Zhang et al. developed a multispectral rapid response temperature measurement system grounded in the principles of multispectral radiation thermometry. This system employs multiple optoelectronic detectors and has been validated for its accuracy in high-temperature measurements [16]. Zhentao Wang et al. developed a multispectral thermal imager designed for measuring temperature fields induced by laser damage. The imager incorporates a five-channel filter photometer with four operational wavelengths and a dedicated calibration channel, facilitating accurate data acquisition across multiple spectral bands. By employing multispectral temperature measurement techniques, we can invert the spectral data to acquire the

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respective temperature fields. The measurement error, when compared to temperatures measured using thermocouples, was found to be less than 1.5 % [2]. Hairi Huang et al. conducted a comprehensive review of data processing algorithms employed in multispectral radiation thermometry, encompassing key factors such as measurement accuracy, processing time efficiency, and modelling methodologies. Notably, they emphasized the prevalence of emissivity assumption models in multispectral thermometry and their successful application in diverse measurement scenarios involving various objects [17]. Currently, spectral emissivity estimation heavily depends on assumed models. When these models closely match real-world emissivity models, temperature inversion and emissivity determination demonstrate high accuracy. Significant discrepancies occur between the inverted results and actual conditions when there is a mismatch between the model and reality [18]. Emissivity assumption models in spectral thermometry highlight constraints in temperature measurement of complex materials or those experiencing dynamic property changes during combustion processes. Additionally, this method suffers from prolonged iterative solution times and a lack of overarching theoretical guidance in choosing suitable emissivity models. Hence, optimizing the emissivity model in multispectral thermometry is essential to acquire richer and more detailed surface information of the sample under measurement.

In this study, aluminium alloy and stainless steel, commonly used in construction, aerospace, high-end equipment manufacturing, and other industries, were chosen as the research materials [19–25]. Radiation spectroscopic investigations were conducted on the damaged area resulting from laser interaction with aluminium alloys and stainless steel. The multispectral thermometry technique beneath the damaged area was refined to deduce the presence of ablation phenomena by analyzing temperature and spectral emissivity variations obtained through multispectral thermometry. This optimization approach is anticipated to facilitate a comprehensive examination of the spatial distribution and alterations within the laser-induced damage zone.

2. Theoretical model

In practical scenarios of ablation temperature measurement, the spatial temperature distribution across the target is non-uniform, particularly under low heat transfer coefficients. This leads to potentially significant local temperature gradients, resulting in variations in radiation spectral line shapes in overlapping regions with different temperatures. Following the principle of multispectral temperature measurement, the impact will manifest as a linear shift in spectral emissivity. The straightforward method of temperature zoning, which allocates varying weights to these temperature zones [15], lacks precise quantitative criteria for weight determination in the presence of ablation or other nonlinear influences. To address this problem, the combination of spatial weighting and emissivity can be utilized as a crucial research parameter, defining a 'Spatially Weighted Emissivity' (SWE), as shown in Equation (1).

$$\varepsilon_\Gamma(\lambda, T) = \mathrm{norm}\left(\frac{S(\lambda, T)}{M_b(\lambda, T)}\right) \tag{1}$$

Wherein (λ,) denotes the actual spectral data that has undergone radiant intensity calibration. (λ,T) represents the ideal blackbody spectral radiation distribution, namely,

$$M_b(\lambda, T) = \frac{\varepsilon_0 c_1}{\lambda^5\left(e^{\frac{c_2}{\lambda T}} - 1\right)} \tag{2}$$

Here, ε_0 represents the spectral emissivity of an ideal blackbody, where $\varepsilon_0 = 1.0$, λ stands for the wavelength, c_1 is the first radiation constant defined as $c_1 = 2\pi hc^2 = 3.74 \times 10–16$ W•m^2, where h is Planck's constant (h = 6.63×10^{-34} J•s), and c is the speed of light (c = 3×108 m/s), c_2 is the second radiation constant, expressed as $c_2 = h \bullet c/k_B = 1.43 \times 10^{-2}$ m•K, with k_B denoting the Boltzmann constant (k_B = 1.38×10^{-23} J/K), and T representing temperature. To simplify, the spatially weighted emissivity is normalized by referencing the maximum value within a specific spectral interval (with a unit of 1) to scale the ratio of S to M_b.

When the irradiated area and the detection zone within the spectrometer's entrance aperture exhibit a wide temperature distribution, the spectral emissivity function ε_Γ(λ,T) is affected by the temperature distribution across the heated sample region. In this scenario, if we assume the maximum temperature within the detection zone is T_{max}, the radial temperature distribution of the radiating area can be described by equation (3).

$$T(r) = g(T_{\max}, r) \tag{3}$$

Within the detection aperture radius r_a, the total spectral radiation distribution can be represented as follows:

$$M(\lambda) = \frac{\int_{T_{\min}}^{T_{\max}} M_b(\lambda, T)dT}{T_{\max}} = \int_0^{r_a} \frac{c_1 \varepsilon(\lambda)}{\lambda^5\left(e^{\frac{c_2}{\lambda g(T_{\max}, r)}} - 1\right)} r dr / (r_a^2) \tag{4}$$

When the temperature gradient in the mentioned interval is small (i.e., minimal temperature variation), the function resulting from the integration of $M(\lambda)$ tends to align with the radiant emittance distribution function. Theoretically, ε_Γ(λ, T) would then simplify to the following expression, where c int represents the integral constant.

$$\varepsilon_\Gamma(\lambda, T) \propto \frac{M}{M_b} = \frac{\frac{c_{\mathrm{int}} c_1 \varepsilon(\lambda)}{\lambda^5\left(e^{\frac{c_2}{\lambda T_{\max}}} - 1\right)}}{\frac{c_1}{\lambda^5\left(e^{\frac{c_2}{\lambda T_{\max}}} - 1\right)}} \propto \varepsilon(\lambda) \tag{5}$$

When calculating M or measuring temperature using multispectral thermometry, the emissivity $\varepsilon(\lambda)$ is typically assumed to exhibit a smooth variation with wavelength. Additionally, due to the necessity for emissivity to be bounded within physical constraints, especially in the context of metals, it shows minimal fluctuations in the visible-near-infrared wavelength range. As a result, the emissivity $\varepsilon(\lambda)$ can be approximated using a polynomial expression, with higher-order terms often deemed insignificant.

$$\sum_{i=0}^{n} \alpha_i \lambda^i, n = 1 \sim 3 \tag{6}$$

Assuming that the emissivity ε(λ) follows a nearly linear form (first-order), it is evident that the calculated ε_Γ (λ, T) from the actual spectral data S(λ, T) will show a strong linear correlation with S (λ, T). On the other hand, a significant temperature gradient within the detection range leads to a notable impact of the spatial temperature distribution on the shape of (λ), thereby diminishing its linear correlation with the blackbody radiation equation.

Therefore, it is deduced that a decrease in the temperature spatial gradient within the detection area leads to an increase in the cross-correlation coefficient between ε_Γ (λ, T) and S (λ, T). A reduced temperature spatial gradient typically indicates either a sudden drop in the energy coupling efficiency of strongly irradiated regions or a sudden rise in local specific heat capacity, reflecting the influence of latent heat due to phase transitions. These two scenarios demonstrate significant nonlinear effects and the initiation of phase changes, both identifiable by distinct spectral signatures. Therefore, integrating spatially weighted emissivity ε_Γ into multispectral temperature measurement techniques aids in determining whether the target subjected to intense laser irradiation has experienced ablation.

3. Setup of the experiment

To investigate the ablation process of aluminum alloy (2A12) and stainless steel (SUS304) under continuous 1070 nm laser irradiation and elucidate the damage mechanisms involved, a comprehensive characterization of the laser-irradiated and ablated samples was performed using spectral thermometry, infrared thermography, monitoring light transmission through the central hole, and visible light imaging techniques. The study also aimed to validate the rationale for incorporating spatially weighted emissivity $\varepsilon\Gamma$ in the multispectral thermometry approach. We set up the experimental testing platform shown in Fig. 1. A continuous 1070 nm laser (SMATLas 4S-3000 W) was directed onto the surfaces of aluminum alloy/stainless steel specimens. Ablation occurs when the material absorbs a critical level of laser energy, sometimes leading to flame generation, which can be considered as low-temperature plasma. The complete damage process was monitored and recorded simultaneously using various instruments: a thermal camera (FLIR T640), a spectrometer (AvaSpec-ULS2048XL-EVO-UA fibre optic spectrometer), laser power fluctuation measurements (FL1100A-BB-65), and a visible light camera (MV-CS050-10GM CMOS). The diameter of the laser spot on the target in the experiment was determined to be 2.6 mm using the knife-edge method.

A 1-mm-diameter hole was precisely machined at the specimen's centre. The power passing through this hole was continuously monitored using a power meter. Any deviation in the power meter readings may suggest hole deformation from ablation or the initiation of a plasma shielding effect, impacting power transmission. Prior to the experiments, the AvaSpec-ULS2048XL-EVO-UA spectrometer underwent calibration with a tungsten filament standard lamp (model LISL-100) from Hangzhou Ulike Technology Co., Ltd. This initial calibration guarantees precise spectral measurements during all subsequent tests.

During the initial phase of the experiment, we employed a setup comprising two cascaded filters with central wavelengths of 515 nm and 1000 nm. However, we observed a significant glare effect in the sample's field of view, which manifested immediately upon laser activation. Our analysis suggests that this effect is primarily due to the leakage of pump light spectrum around 970 nm from the laser. Additionally, as the laser operated continuously, the thermal radiation spectrum emitted by the ablated sample intensified, causing significant interference with the camera. To tackle this issue, our research team upgraded the filter setup by incorporating two additional short-pass filters with central wavelengths of 700 nm and 800 nm into the cage cube placed in front of the lens. This enhancement ensured attenuation levels exceeding OD10 above 800 nm and above OD4 above 515 nm, effectively eliminating undesired wavelengths. The introduction of this cascaded short-pass filter configuration enhanced image visibility for direct observation.

4. Results and Discussion

4.1. The analysis of ablation spectral

4.1.1. Spectral unmixing techniques

Based on the experimental outcomes shown in Fig. 1, we obtained spectral and image data from the laser ablation processes of aluminium alloy and stainless steel. Through meticulous analysis of this dataset, we aimed to uncover the mechanisms of laser ablation in aluminium alloy and stainless steel, revealing the complex dynamics in their respective ablation processes. Initially, multispectral thermometry was used in spectroscopic analysis to study the temperature field distribution characteristics during ablation. Spectral decoupling is necessary to isolate the continuous spectrum for temperature measurement, involving the separation of discrete spectral components from the continuous background. The following detailed steps were taken for this purpose.

The measured spectra were first subjected to standard calibration procedures, which included background subtraction, wavelength calibration, and blackbody radiation calibration. Subsequently, the Grubbs criterion was used to determine the peak positions and approximate wavelength ranges for discrete peaks. At the initial stage of laser activation, with the target temperature still low, the continuous spectrum was nearly absent. The spectra during this period provided initial values for nonlinear parameters like line shape and linewidth of the discrete peaks. Based on the fitting results, the Lorentzian profile effectively described the peaks of the pump laser spectrum at around 970 nm and the laser emission peaks at around 1070 nm, both with line widths of about 10 nm. To ensure comparable intensities of both the pump and emission spectrum peaks in the acquired spectra, a short-pass filter with a central wavelength of 1000 nm (OD4) was placed in front of the spectrometer probe. The intensities of the three spectral components were determined using the matrix reconstruction method, a mainstream global fitting algorithm for transient absorption spectroscopy. The

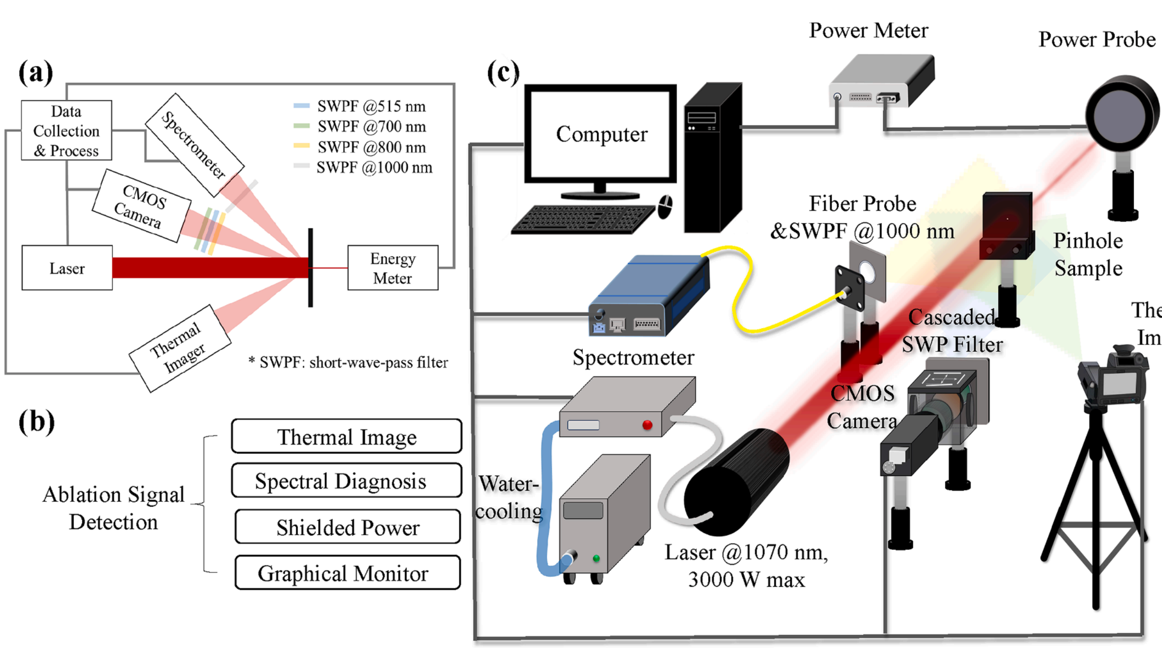

Fig. 1. The experimental testing platform includes (a) A schematic diagram illustrating the interconnection of optical instruments for laser delivery and signal detection. (b) A logic tier diagram of the ablation signal monitoring system shows the hierarchical structure of data acquisition and processing steps for tracking and managing ablation-related signals. (c) A 3D simulation model diagram of the optical components and instruments in the experimental system provides a virtual representation of the layout and arrangement of key elements like lasers, spectrometers, and imaging devices in the setup.

comparison of component intensities and fitted profiles is shown in Fig. 2: The grey-blue line represents the thermal radiation spectrum, the pink line represents the 970 nm pump laser spectrum, the yellow line represents the laser scattering spectrum, and the grey line represents the reconstructed spectrum. The dashed vertical line indicates the cutoff at 1000 nm (OD4), while the small light-grey dots scattered across the graph depict the initial raw spectral data points.

In Fig. 2, the change in parameters during ablation is inconsistent between stainless steel and aluminium alloy. In aluminium alloy, the peak intensity of the discrete spectrum can indicate ablation occurrence, as seen in Fig. 2C. After approximately 80 s of irradiation, the peak of the discrete spectrum becomes unstable, marking the end of the "plateau period" due to alterations in the target material's surface shape and reflectivity. When metallic aluminium melts within the irradiated area of the aluminium alloy target, local deformation occurs due to gravity. Although immediate morphological changes are minimal because of the adherence of the high-melting-point oxide film, the reflected light's phase is susceptible to surface shape variations. Even slight local surface deformations can significantly alter the intensity of corresponding characteristic peaks, especially for peaks further from the target, where greater distances result in higher peak intensities. Additionally, the aluminium alloy's low surface absorption and high heat transfer coefficient lead to poor laser coupling efficiency, causing delayed thermal radiation spectrum appearance and hindering accurate multispectral temperature measurements.

After 136 s of ablation, a sudden increase in the thermal radiation spectrum intensity is observed for the aluminium alloy target, accompanied by a sharp decrease in the intensity of discrete spectra. Observations from the target camera confirm that this phenomenon corresponds to the internal liquefaction of the target, resulting in the breakdown of the material structure in the irradiated area and potential leakage. When molten aluminium is suddenly exposed to atmospheric oxygen, it undergoes a vigorous oxidation reaction, leading to high temperatures. Prior to this point ($\sim$120 s), a stable and significant continuous spectrum suitable for multispectral thermometry only appears on the short-wave side at 1000 nm. This delay highlights the challenge of locally accumulating thermal energy within the radiation zone of the aluminium alloy before breakdown due to its high latent heat of fusion, thermal conductivity, and low surface absorptivity compared to other metals. Under the current experimental conditions (limited by the maximum power of 1100 W that the available power meter can handle), ablating aluminium alloy proves to be particularly difficult, requiring over 80 s compared to stainless steel, emphasizing the complexities of the ablation process influenced by the unique thermophysical properties of the material.

In contrast to aluminium alloy, stainless steel exhibits a rapid temperature increase within 2 s and undergoes ablation after about 3.5 s of exposure to a lower irradiation power of 520 W. As shown in Fig. 2 (a) and (b), among the three separate spectral components, the discrete spectrum intensity does not show a "plateau phase," indicating that changes in the continuous spectrum intensity at the beginning of ablation are not significant. Furthermore, in many real-world scenarios where prior knowledge about the target material's specific characteristics, such as melting and boiling points, is lacking, especially for non-cooperative targets, it would be unwise to assume that damage occurs once the target temperature exceeds a certain threshold. The intricate ablation dynamics in stainless steel, characterized by its quick response to laser exposure and the lack of a clear, steady state in spectral emissions, highlight the importance of meticulous spectral analysis and customized approaches for comprehending and predicting material behaviour during laser ablation.

4.1.2. Spatially weighted emissivity-based multispectral thermometry method

Our research team utilized multispectral thermometry along with a novel weighted radiative spectrum inversion method to extract variations in temperature and emissivity distributions to tackle this challenge. This approach allows for determining target ablation and shows potential as a practical and feasible solution. Along with theoretical simulation results, the spatial distribution of temperature affects the shape of blackbody radiation spectra, potentially providing strong support for evaluating target heating and ablation scenarios. Most importantly, extracting temperature fluctuations and spectral emissivity changes using multispectral thermometry offers a way to deduce the

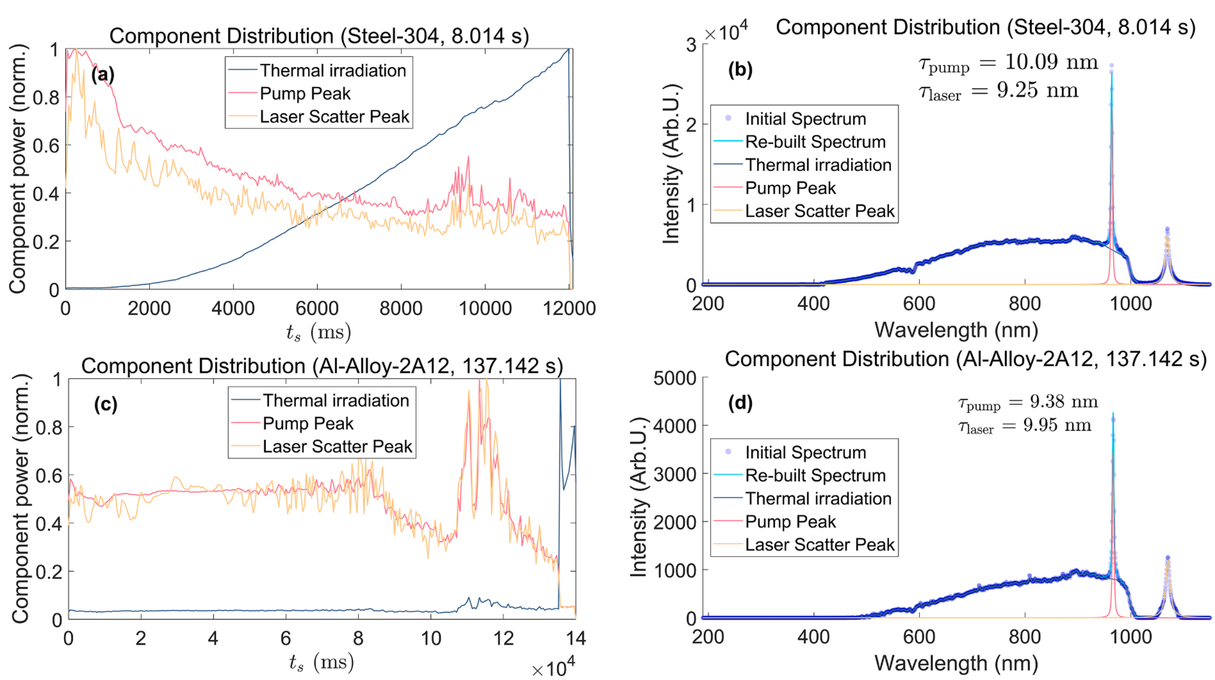

Fig. 2. The weighting of discrete and continuous spectral components and their corresponding fitted profiles were analyzed. (a) Evolution of component weighting over ablation time for the steel-304 sample; (b) Component profiles, initial spectrum, and reconstructed spectrum of the steel-304 sample; (c) Evolution of component weighting over ablation time for the Al-Alloy-2A12 sample; (d) Component profiles, initial spectrum, and reconstructed spectrum of the Al-Alloy-2A12 sample.

initiation and progression of ablation processes. This integrated methodology leads to improved comprehension and real-time monitoring of laser-material interactions, thus enhancing precision and control in laser-based material processing applications.

The multispectral thermometry results for the stainless steel target under laser irradiation at 2 s and 8 s are shown in Figs. 3 and 4, respectively. It is clear that prior to reaching the melting point of the target material, the spatially weighted emissivity of the target (depicted in red) is significantly affected by the temperature gradient across its surface, leading to a notable deviation from the spectral line shape (shown in gray). In contrast, the spatially weighted emissivity profile closely matches the spectral line shape for the ablated sample, indicating a more uniform temperature distribution and a potential alteration in surface characteristics post-ablation. This finding highlights the dynamic changes in emissivity patterns during laser ablation, influenced by temperature increase and material transformation.

4.1.3. Weighted radiative spectrum inversion method

In terms of temperature computation, we utilized two methodologies: the first is the widely adopted multispectral thermometry, and the second is the proposed weighted radiative spectrum inversion technique. The motivation for introducing the latter method arises from the limitations faced with conventional multispectral thermometry in scenarios where the signal-to-noise ratio of the thermal signal is low at low target temperatures or in specific temperature ranges where the spectral intensity gradient within the selected bands is too tiny to capture the nonlinear characteristics of thermal radiation spectra adequately. In such situations, the accuracy and stability of temperature calculations using multispectral thermometry cannot be reliably ensured. Importantly, computations may result in negative temperature values at lower temperatures, indicating non-convergence issues. The critical distinction between the weighted radiative spectrum inversion method and traditional multispectral thermometry lies in assuming that the spectral emissivity of the sample remains approximately constant across the selected spectral bands and is unaffected by spatial temperature variations on the target surface. With this assumption, the approach proceeds with the following considerations.

$$\frac{S(\lambda, T)}{M_b(\lambda, T)} \approx C \tag{7}$$

Ordinary least squares regression, relying on the sum of squared absolute errors, inadvertently gives excessive weight to wavelengths with stronger signals, especially those on the longer wavelength side. This imbalance hinders the extraction of the intrinsic nonlinear features present in the thermal radiation spectrum, features that arise from temperature fluctuations. As a result, optimization strategies based on least squares methods inadequately capture the nonlinear parameters that characterize the behaviour of the thermal radiation spectrum.

To ensure the meaningful and optimal contribution of weaker signal data in the optimization process, it is necessary to formulate alternative objective functions to replace the conventional Euclidean distance metric in the data domain. This strategic adjustment aims to promote a fairer contribution of all spectral components, thereby improving the extraction of temperature-dependent nonlinear characteristics from complex spectral patterns, especially when the thermal irradiation peak occurs at wavelengths within the detection range. Adjustments in two aspects of weight factors are crucial to meet these needs. Firstly, the objective function should devalue linear components and focus on the sum of squared relative errors to highlight the nonlinear traits. Secondly, to prevent an excessive emphasis on weak signals, especially those close to the signal-to-noise ratio (SNR) limits, a weighting scheme proportional to signal strength should be integrated with the squared relative errors of the respective data points. Therefore, a proposed formulation could be presented as follows:

$$R_i = \frac{S(\lambda_i, T)}{M_b(\lambda_i, T)} \tag{8}$$

The objective function can be designed as follows:

$$D = \sum_{i=1}^{n} \left(\frac{R}{\bar{R}} - 1\right)^2 \cdot S(\lambda_i, T) \tag{9}$$

Moreover, utilizing this objective function allows for the application of traditional optimization algorithms like the conjugate gradient method. The computational outcomes from the two temperature measurement methods are illustrated in Fig. 5 below. The left plot displays the results of multispectral thermometry in pink and weighted radiative spectrum inversion in red. The blue and red lines represent the same data. Still, they are colour-coded to emphasize areas of high-temperature gradient (blue for heating state) and low-temperature gradient (red for ablation state). The superiority of weighted radiative spectrum inversion over multispectral thermometry is apparent in terms of convergence in weak signal regions and enhanced measurement accuracy around 1500 °C, aligning closely with infrared thermography results (refer to the subsequent section). Hence, it acts as a valuable supplement to multispectral thermometry.

However, to improve the stability and convergence of the temperature measurement model, this approach assumes that the target's emissivity remains constant and ignores spatial variations in emissivity caused by temperature gradients. As a result, the calculated temperature trend appears relatively smooth, making it difficult to determine the

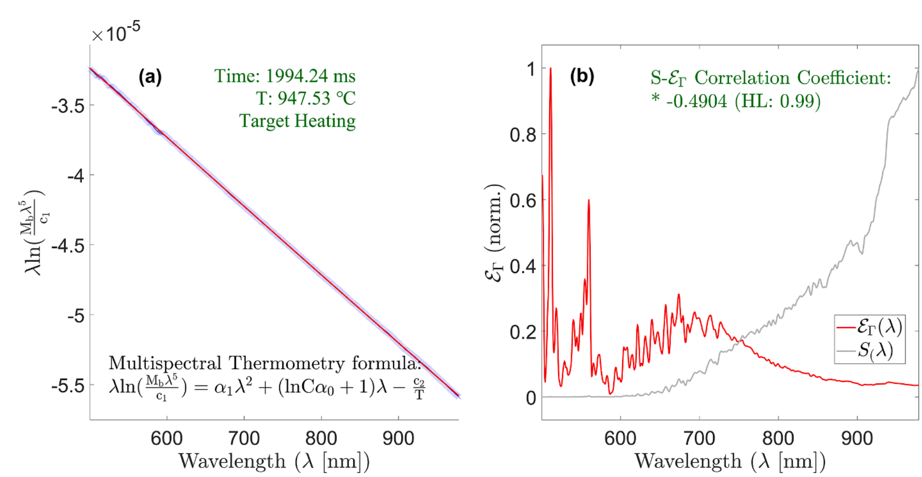

Fig. 3. Multispectral thermometry curve of stainless steel when subjected to laser irradiation for 2 s. (a) Multispectral thermometry fitting curve; (b) the $\varepsilon_\Gamma(\lambda, T)$ of SWE and the initial spectrum $S(\lambda,T)$.

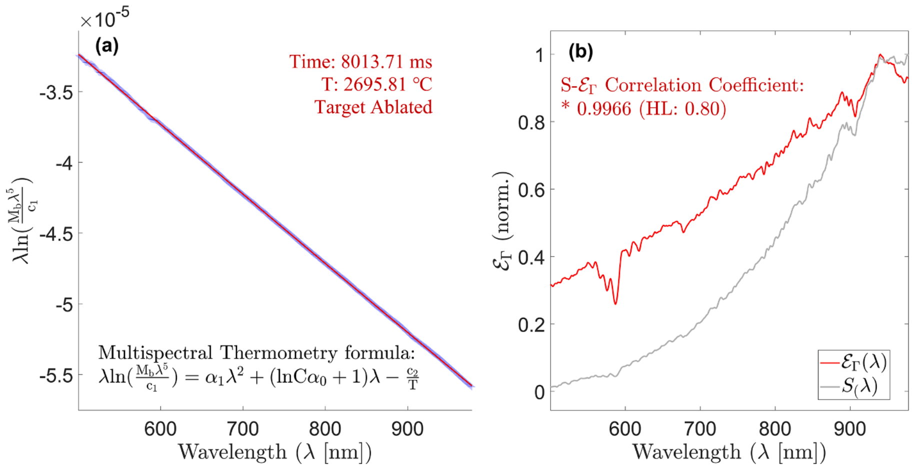

Fig. 4. Multispectral thermometry curve of stainless steel under laser irradiation for 8 s. (a) Multispectral thermometry fitting line; (b) the $\varepsilon_\Gamma(\lambda, T)$ of SWE and the initial spectrum $S(\lambda,T)$.

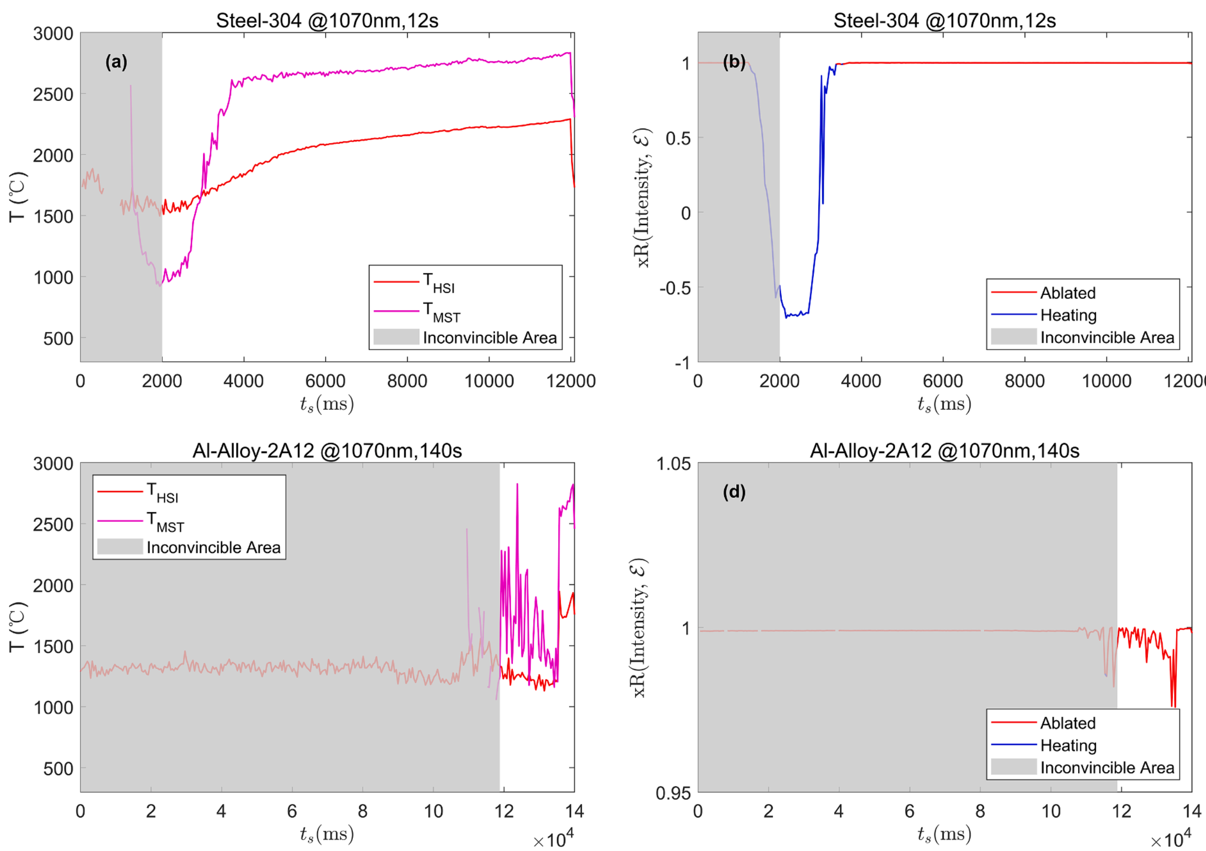

Fig. 5. Temperature calculation results using two thermometry approaches, the weighted radiative spectrum inversion method (HSI) and multispectral thermometry (MST), are presented. The two sub-figures on the left show temperature evolution curves obtained by HSI and MST for the steel-304 sample (a) and the Al-Alloy-2A12 sample (b). The grey areas are identified as inconclusive due to abnormal trends or values outside the range of [0,3000] ℃. The two sub-figures on the right display the evolution curves of the ε_Γ-S correlation coefficients for the steel-304 sample (c) and the Al-Alloy-2A12 sample (d).

precise moment of sample damage. In the case of aluminium alloys, both temperature measurement methods perform poorly due to the combined effects of a dense oxide layer on the surface, high thermal conductivity, and low absorptivity. Instead, conducting post-irradiation image analysis of the sample, focusing on macroscopic ablation characteristics such as target trajectory, deformation, ablation plume, and smoke, can serve as evaluation indicators.

4.1.4. Comparative validation of the weighted radiative spectrum inversion method

When utilizing multispectral thermometry, the significantly different melting/vaporization temperatures of various metals present challenges. Specifically, in the case of aluminium alloys, it is challenging to acquire high signal-to-noise ratio data in the visible spectrum before the onset of “liquid leakage” for reliable multispectral temperature measurement. Therefore, reference temperatures are obtained using a trusted infrared thermal imager. Fig. 6 illustrates the temporal temperature

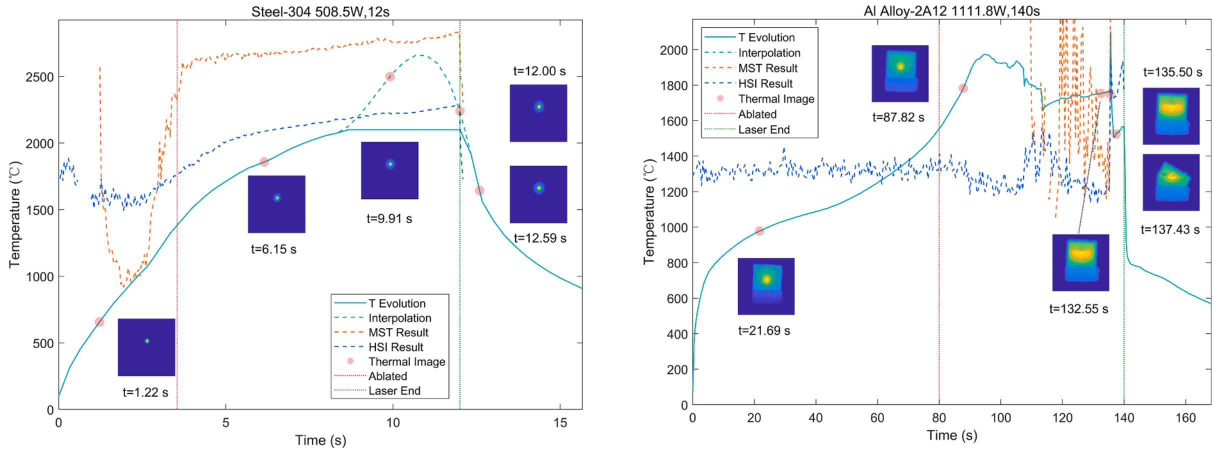

Fig. 6. Comparative temperature measurement using 3 methods including HSI, MST and thermal image for Steel-304 (a) and Al-Alloy-2A12 (b).

trend at the laser spot centre, which represents the highest temperature point recorded by the infrared thermal imager (with emissivity set to 0.30 for stainless steel and 0.28 for aluminium alloy as per the FLIR manual [26]). It also includes pseudo-colour maps showing temperature distributions at various randomly selected instances. Comparison of these results with temperatures obtained through multispectral thermometry (MST) and the weighted radiative spectrum inversion method (HSI) revealed that in high-temperature conditions (>1500 °C), the HSI method demonstrates superior stability and accuracy in temperature calculations compared to MST. Moreover, the HSI outcomes closely correspond to those measured by the infrared thermal imager.

Meanwhile, the temperature data from the infrared thermography supports our analysis of spatial weighting and heat transfer. It confirms that aluminium alloy conducts heat faster than stainless steel, leading to higher temperatures at the sample edges and holder during heating, a phenomenon not seen with stainless steel.

To prevent damage to the spectrometer equipment, a collecting optical system was not placed in front of the spectrometer fibre optic probe. Consequently, the probe can only target a small area near the centre of the irradiation zone. In such cases, a steep temperature gradient within this limited region significantly affects the correlation between spatially weighted emissivity and spectral linearity. For stainless steel, characterized by relatively slow thermal conductivity, the initial heating stages exhibit a substantial temperature gradient in this area. Only as the temperature gradient diminishes, influenced by phase transitions like ablation and other factors, does the correlation coefficient between spatially weighted emissivity and the spectrum increase. This enhancement enables the diagnosis of ablation onset in the sample by analyzing the linear correlation between spatially weighted emissivity and the spectrum.

Conversely, in aluminium alloy, the temperature gradient in the central region remains consistently low. The thermal radiation signal received by the spectrometer in this study is limited in effectiveness, maintaining consistently high correlation coefficients and resulting in ineffective diagnosis. However, ablation diagnosis can still be accomplished by monitoring trends in discrete spectrum intensity changes caused by pump light and the laser (as shown in Fig. 2 (c)), effectively compensating for the inability to diagnose ablation based on spatially

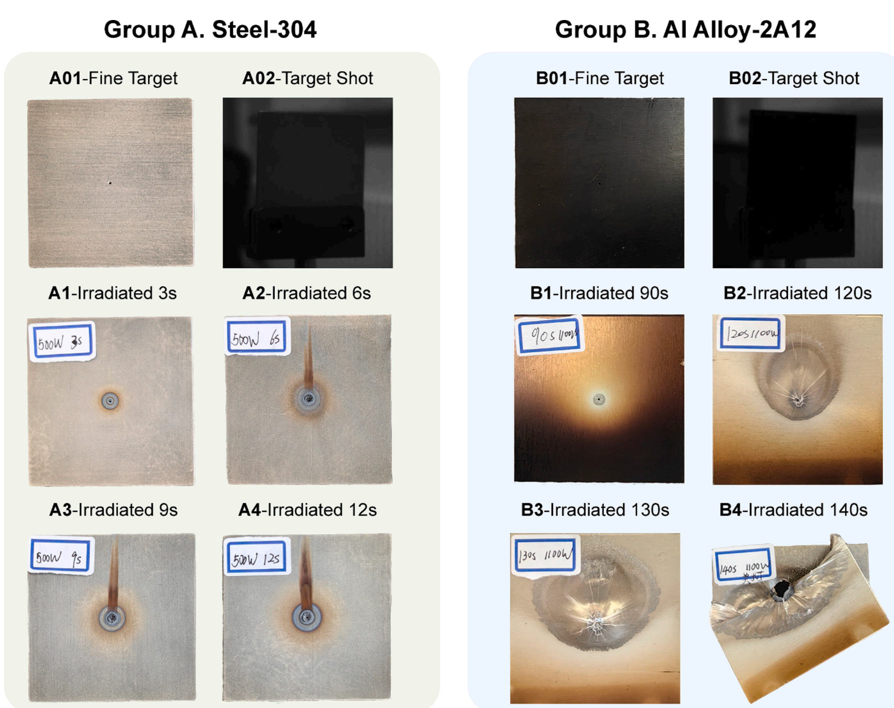

Fig. 7. Damage morphologies of stainless steel (Group A) and aluminium alloy (Group B) under continuous laser irradiation at different durations.

weighted emissivity.

4.2. Analysis of damage morphology

The aluminium alloy samples in the experiment underwent surface oxidation to turn black, enhancing their surface absorptivity. However, this surface layer is heated and removed before reaching the temperature/timepoint required for ablation (as shown in Fig. 7B1), ensuring the thermodynamic state of the aluminium alloy before and after ablation remains consistent. The original condition of the samples (A01/B01), their appearance in the camera's field of view (A02/B02), and their states after continuous irradiation for specific durations (A1-4/B1-4) were recorded. The ablation processes of the two samples show significant differences due to their distinct physical mechanisms.

Stainless steel undergoes a whole process of melting, vaporization, and ejection when exposed to laser irradiation, as confirmed by comparisons with theoretical models and simulation results [21–23]. An ablation pit overflows at the sample's centre with prolonged laser exposure. Additionally, in Fig. 7 A2-A4, brownish-black traces are observed surrounding and above the ablation pit, with the dark band above the pit progressively intensifying. This occurrence is explained by the oxidation of metal vapour rising during ablation, which reacts with atmospheric oxygen to produce oxide particles. Subsequently, these particles cool and settle above the sample, resulting in the formation of brownish-black marks. Conversely, aluminium alloy gradually heats up when exposed to laser irradiation, leading to deformation as the internal metal melts. Nevertheless, this deformation is less pronounced because of the swift oxidation of aluminium upon contact with air, resulting in the formation of a dense oxide layer that encloses the molten metal.

With prolonged laser irradiation, the high heat transfer efficiency and latent heat of aluminium alloys in solid and liquid states cause the solid metal around the ablation centre to absorb energy from the laser, leading to liquefaction continuously. The liquefied metal then moves downward due to gravity, forming a bulge. Simultaneously, thermal convection occurs in the molten metal, causing a faster melting rate at the top and creating an elliptical molten region slightly off-centre upwards. As the molten region reaches the sample's edge, heat accumulation increases, potentially causing partial vaporization or other thermodynamic effects in the central area. This process can result in expansion, rupture of the oxide film, leakage, severe oxidation, breakage, and structural disintegration due to gravitational forces.

4.3. Variations in laser coupling efficiency during the ablation process

The research group further investigated the variations in laser coupling efficiency during sample ablation. Initial insights were obtained by analyzing experimental videos and the evolution of laser power leakage through the small aperture, as shown in Fig. 8(b). It was confirmed that, without additional surface coatings, aluminium alloys face challenges in generating low-temperature plasmas (similar to flames) on the target surface. Additionally, high-temperature plasmas, like those from short-pulse laser breakdown spectroscopy, are largely absent due to insufficient electron density at the surface. Approximately 80 s after the initial metal melting, changes in the aperture's position and shape resulted in a slight decrease in the transmitted laser power.

To explore the physical mechanisms underlying this decrease, the research team conducted additional processing and analysis of images taken during laser irradiation of the targets. Initially, from a spectroscopic standpoint, the effects on target image pattern recognition can be ascribed to two primary factors: (i) laser-induced influences, particularly scattered signals from the pump and emitted lasers across the long-visible to short-near-infrared spectral range, and (ii) extraneous radiation from the sample itself, including metal combustion luminescence and stimulated emission signals. These signals may disrupt imaging equipment, leading to a decline in image recognition precision.

Possible physical processes that generate these harmful radiation signals (at shorter wavelengths) include thermal radiation, metallic flame colouration (emission spectrum of excited metal atoms), excitation fluorescence, or upconversion from doped or coated semiconductor/dye materials, and anti-Stokes or stimulated anti-Stokes scattering. Such radiations are rarely detected at wavelengths below 500 nm. Therefore, employing a set of short-pass filters allows for capturing clearer images.

All synchronized experiments were conducted in a dark environment during formal experiments to eliminate the influence of ambient light on the spectrometer. Fig. 9 displays the integrated grayscale value variation curves of the laser spots over time and snapshot images taken at specific time points.

In Fig. 9(a), the aluminium alloy target disintegrates around 130 s of laser exposure. Subsequently, liquid metal leaks from the cooler upper part of the sample undergoes oxidation upon irradiation, and generates significant thermal radiation. In Fig. 9(b), stainless steel, when subjected to laser irradiation, emits abundant thermal radiation in the ablation region due to the ejection and oxidation of vaporized metal. By observing the decreasing trend in leaked laser power through the aperture (Fig. 8) and comparing it with the sample image after 12 s of ablation, a consistent decline trend is evident, as shown by the light green line in Fig. 9(a), starting from the ablation onset at 2 s. Additionally, the morphology of the ablation pit in Fig. 7(A1) indicates minimal changes in the size and shape of the central aperture, implying reduced influence from liquid metal. This trend could be attributed to the expansion of gas clouds formed by vaporized metal, resembling flames and potentially creating a low-density, low-temperature plasma. Despite the adverse impact of these expanding gases on subsequent laser

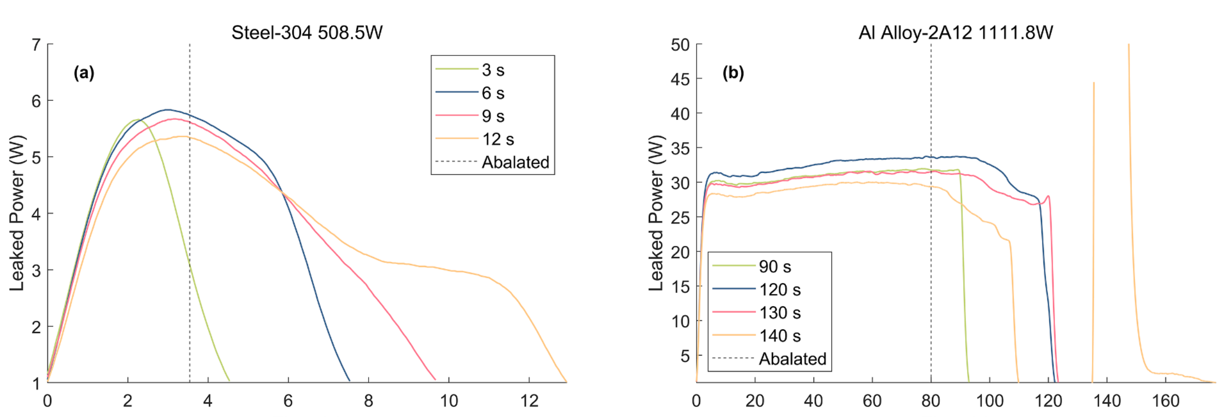

Fig. 8. The Evolution of laser power transmitted through the aperture is depicted. The curves showing the power leakage over different durations are presented for (a) steel-304 and (b) Al-Alloy-2A1.

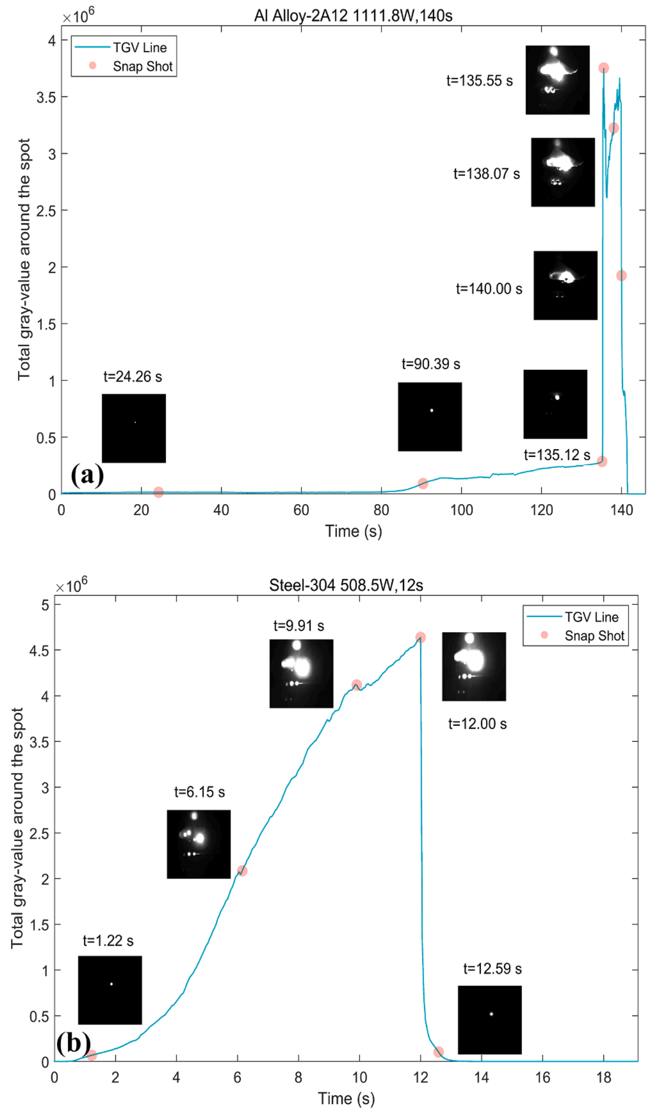

Fig. 9. Total grey-value curves and images captured by the CMOS camera of laser-irradiated aluminium alloy (top) and stainless steel (bottom).

irradiation, vaporization has already taken place by this stage, rendering ablation unavoidable.

The emergence of this phenomenon positively influences future research. Coating the target surface with materials that vaporize easily and absorb heat can lead to the expansion of low-temperature combustion gases, potentially degrading laser beam collimation. This degradation reduces the coupling efficiency between the laser and the target but ensures efficient absorption of deposited energy. The research team plans to conduct additional exploratory work to build upon these findings and improve protective strategies against laser irradiation.

5. Conclusion

In this study, the research group developed a multidimensional platform for temperature measurement and remote ablation characteristic diagnosis. Through a comparison of three measurement techniques (spectral thermometry (MST), weighted radiative spectrum inversion method (HSI), and infrared thermography), we determined that the traditional MST technique is more effective for ablation diagnosis in terms of spatial weight emissivity and spectral correlation analysis compared to the other two methods. In the low-temperature range ($<$ 1500℃), HSI and thermal imaging temperature measurements exhibit similar results, with greater consistency in HIS temperature measurements. Therefore, the combination of these three methods results in more precise temperature measurement and ablation diagnosis than using a single process. Additionally, by integrating data from central perforation light leakage monitoring and imaging observation methods, the ablation mechanisms of aluminium alloy and stainless steel were successfully elucidated. The findings of this study are essential for evaluating laser damage effects, analyzing material properties, and optimizing temperature measurement techniques.

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CRediT authorship contribution statement

Jiamin Wang: Writing – original draft, Methodology, Investigation, Data curation. **Yunfeng Zhang:** Writing – review & editing, Software, Methodology, Investigation, Data curation, Conceptualization. **Changbin Zheng:** Supervision, Methodology, Conceptualization. **Kuo Zhang:** Validation, Formal analysis, Data curation, Conceptualization. **Junfeng Shao:** Visualization, Validation, Supervision, Data curation, Conceptualization. **Chunrui Wang:** Resources, Methodology, Funding acquisition. **Yunzhe Wang:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fei Chen:** Writing – review & editing, Supervision, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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